

JACOB COOK

SIEM Engineer II | Detection Engineering & Log Onboarding | Sigma · Wazuh · KQL · Python · Pipeline Tuning

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SIEM-focused security engineer with hands-on experience standing up a full Wazuh stack, onboarding log sources, normalizing telemetry, and authoring detection content mapped to MITRE ATT&CK. Strong in Sigma rule development, dashboard/alert tuning, and reducing data-quality noise to support SOC operations and threat hunting. Python, Bash, SQL, and KQL/Sigma queries used daily in lab work. CompTIA-certified across four disciplines (Security+, CySA+, PenTest+, SecurityX) with BS Computer Science and MS Cybersecurity in progress.

TECHNICAL SKILLS

SIEM & Detection Engineering: Wazuh SIEM (build & admin), Sigma Rule Authoring, MITRE ATT&CK, Correlation/Alert Logic, Detection Tuning, KQL (familiar), SPL (familiar), Dashboards & Reporting, Log Source Onboarding

Pipelines & Logging: Log Normalization, Syslog/API/Agent Ingestion, Data Quality & Routing (Cribl-style ETL concepts), Microsoft Sentinel (familiar)

Cloud & Identity: Azure AD / Entra ID, Okta, AWS labs, SSO / RBAC, Terraform

Development: Python, Bash, SQL, Regex, FastAPI, Git

TECHNICAL PROJECTS

Stryker / Intune Detection Pack: Detection-as-Code

2026

- Authored 6 Sigma-format correlation rules for Intune MDM abuse mapped to MITRE ATT&CK, with enrichment logic that eliminated 100% of false positives in a 100-event simulation (80 benign suppressed, 20 malicious caught) using KV-style admin baseline lookups.
- Documented 3 benign false-positive scenarios per rule (18 total), covering VPN logins, scheduled maintenance, and PIM activations so analysts can tune confidently without dropping known-good telemetry.
- Verification: 43 passing pytest cases in repo (tests/test_rules.py), reproducible against synthetic data.

Blue Team SOC Monitoring Lab: Wazuh SIEM (End-to-End Build)

2026

- Built a full Wazuh SIEM stack from scratch — log ingestion, normalization, dashboards, and alert rules — and validated 12 detection patterns against 242 simulated auth-style events with 100% precision after tuning.
- Reduced false-positive alert volume by 100% by tuning out all 79 benign-only matches (cron jobs, service accounts, scheduled deployments) without dropping a single true positive — the same data-quality discipline platform-side SIEM work demands.
- Verification: 19 passing pytest cases in repo (tests/test_detection_accuracy.py).

SOAR-lite: Incident Response Orchestrator

2026

- Eliminated four manual triage steps (classification, IOC enrichment, notification, ticket creation) by building automated SIEM-triggered playbooks, reducing triage time by over 60% in lab testing.
- Built end-to-end alert automation: webhook ingestion from Wazuh, automatic playbook routing by event type, IP blocking, Slack notification, and incident documentation with zero analyst intervention.
- Verification: 10 passing pytest cases in repo (tests/test_orchestrator.py).

Phishing Analysis Lab

2026

- Triageed 5 real-world phishing samples end-to-end: header inspection, SPF/DKIM/DMARC validation, URL defanging, VirusTotal enrichment, IOC extraction — 100% accurate verdicts against known IOC ground truth.
- Delivered structured triage reports with documented IOCs, confidence assessments, and escalation decisions matching SOC reporting expectations.

G3-GPT: Industry-Sponsored Enterprise AI Platform (Team of 6)

2024

- Secured a production AI platform by engineering RBAC and Azure AD SSO, enforcing identity boundaries that enabled executive-level adoption.
- Shipped enterprise-grade platform on deadline under direct industry sponsor oversight with zero critical security findings at delivery.

PROFESSIONAL EXPERIENCE

Target Distribution Center | Operations Engineer: Systems Monitoring & Incident Response

2019 - 2025

- Maintained 99.5% system uptime by monitoring telemetry across automated sortation infrastructure on rotating shifts — the same continuous-monitoring discipline expected from SIEM platform support.
- Resolved 200+ system anomalies through structured root cause analysis with tracked documentation that reduced repeat incidents by an estimated 30% and supported audit readiness.
- Reduced mean time to recovery by an estimated 20% by coordinating cross-team incident response, including escalation, triage prioritization, and post-incident reporting.

EDUCATION & CERTIFICATIONS

WGU | MS, Cybersecurity (In Progress)

Expected Oct 2026 · GPA 3.50

CSUS | BS, Computer Science

Jan 2025 · GPA 3.50

CompTIA: Security+ · CySA+ · PenTest+ · SecurityX · CSIE (stackable)